

```

MySQL 8.0 Command Line Cli  ×  +  ▾

mysql> INSERT INTO class_info VALUES
  -> (7,'NOIDA'),
  -> (8,'PANIPAT');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

```

```

mysql>
mysql> SELECT *
  -> FROM class
  -> LEFT OUTER JOIN class_info
  -> ON class.id=class_info.id;

```

id	name	id	address
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
4	alex	NULL	NULL
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
3	alex	3	CHENNAI
4	anu	NULL	NULL
5	ashish	NULL	NULL

```

13 rows in set (0.00 sec)

```

```

mysql> SELECT COUNT(*)
  -> FROM
  -> (
  -> SELECT * FROM first_table
  -> UNION ALL
  -> SELECT * FROM second_table
  -> ) A;

```

COUNT(*)
4

```

1 row in set (0.00 sec)

```

JOINS_OPERATIONS_DB_11

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```
MySQL 8.0 Command Line Cli × + v
mysql> SELECT class.name,
->         class_info.address
-> FROM class
-> INNER JOIN class_info
-> ON class.id=class_info.id;
+-----+-----+
| name | address |
+-----+-----+
| abhi | DELHI   |
| abhi | DELHI   |
| adam | MUMBAI  |
| adam | MUMBAI  |
| abhi | DELHI   |
| abhi | DELHI   |
| adam | MUMBAI  |
| adam | MUMBAI  |
| alex | CHENNAI |
| alex | CHENNAI |
+-----+-----+
10 rows in set (0.00 sec)
```

```
MySQL 8.0 Command Line Cli  ×  +  ∨

mysql> SELECT c.id, c.name, ci.address
      -> FROM class c
      -> INNER JOIN class_info ci
      -> ON c.id = ci.id
      -> WHERE c.name = 'adam';
+-----+-----+-----+
| id  | name | address |
+-----+-----+-----+
|  2  | adam | MUMBAI  |
|  2  | adam | MUMBAI  |
|  2  | adam | MUMBAI  |
|  2  | adam | MUMBAI  |
+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT DISTINCT f.name
      -> FROM first_table f
      -> INNER JOIN second_table s
      -> ON f.name = s.name;
+-----+
| name |
+-----+
| adam |
+-----+
1 row in set (0.00 sec)
```

```
MySQL 8.0 Command Line Cli  ×  +  ∨

mysql> SELECT *
      -> FROM class
      -> RIGHT JOIN class_info
      -> ON class.id=class_info.id
      -> WHERE class.id IS NULL;
+-----+-----+-----+-----+
| id  | name | id  | address |
+-----+-----+-----+-----+
| NULL | NULL |  7  | NOIDA   |
| NULL | NULL |  8  | PANIPAT |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT f.name  
-> FROM first_table f  
-> LEFT JOIN second_table s  
-> ON f.name = s.name  
-> WHERE s.name IS NULL;
```

```
+-----+  
| name |  
+-----+  
| abhi |  
+-----+
```

1 row in set (0.00 sec)

MySQL 8.0 Command Line Cli

```
mysql> SELECT *  
-> FROM class  
-> NATURAL JOIN class_info;
```

```
+-----+-----+-----+  
| id  | name | address |  
+-----+-----+-----+  
| 1   | abhi | DELHI   |  
| 1   | abhi | DELHI   |  
| 2   | adam | MUMBAI  |  
| 2   | adam | MUMBAI  |  
| 1   | abhi | DELHI   |  
| 1   | abhi | DELHI   |  
| 2   | adam | MUMBAI  |  
| 2   | adam | MUMBAI  |  
| 3   | alex | CHENNAI |  
| 3   | alex | CHENNAI |  
+-----+-----+-----+
```

10 rows in set (0.00 sec)

```
mysql>  
mysql> INSERT INTO class VALUES  
-> (5, 'ashish');  
Query OK, 1 row affected (0.01 sec)
```

```
MySQL 8.0 Command Line Cli  X  +  v

mysql> SELECT *
-> FROM class
-> CROSS JOIN class_infor;
ERROR 1146 (42S02): Table 'joins.class_infor' doesn't exist
mysql>
mysql> SELECT *
-> FROM class
-> CROSS JOIN class_info;
+-----+-----+-----+-----+
| id  | name | id  | address |
+-----+-----+-----+-----+
| 1  | abhi | 1  | DELHI   |
| 2  | adam | 1  | DELHI   |
| 4  | alex | 1  | DELHI   |
| 1  | abhi | 2  | MUMBAI  |
| 2  | adam | 2  | MUMBAI  |
| 4  | alex | 2  | MUMBAI  |
| 1  | abhi | 3  | CHENNAI |
| 2  | adam | 3  | CHENNAI |
| 4  | alex | 3  | CHENNAI |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)

mysql>
mysql> CREATE TABLE class (
->     id INT,
->     name VARCHAR(30)
-> );
ERROR 1050 (42S01): Table 'class' already exists
mysql> INSERT INTO class VALUES
-> (1,'abhi'),
-> (2,'adam'),
-> (3,'alex'),
-> (4,'anu');
Query OK, 4 rows affected (0.01 sec)
```

```
MySQL 8.0 Command Line Cli  X  +  v

mysql>
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id=class_info.id
-> WHERE class_info.id IS NULL;
+-----+-----+-----+-----+
| id  | name  | id  | address |
+-----+-----+-----+-----+
| 4  | alex  | NULL | NULL    |
| 4  | anu   | NULL | NULL    |
| 5  | ashish | NULL | NULL    |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT f.*
-> FROM first_table f
-> LEFT JOIN second_table s
-> ON f.name = s.name
-> WHERE s.name IS NULL;
```

```
+-----+-----+
| id    | name  |
+-----+-----+
| 1     | abhi  |
+-----+-----+
1 row in set (0.00 sec)
```

```
MySQL 8.0 Command Line Cli  X  +  v

mysql>
mysql> CREATE TABLE first_table(
-> id INT,
-> name VARCHAR(30)
-> );
Query OK, 0 rows affected (0.08 sec)

mysql>
mysql> CREATE TABLE second_table(
-> id INT,
-> name VARCHAR(30)
-> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO first_table VALUES
-> (1,'abhi'),
-> (2,'adam');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO second_table VALUES
-> (2,'adam'),
-> (3,'chester');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM first_table
-> UNION
-> SELECT * FROM second_table;
```

```
mysql> SELECT a.*  
-> FROM first_table a  
-> INNER JOIN second_table b  
-> ON a.id = b.id;
```

```
+-----+-----+  
| id    | name  |  
+-----+-----+  
|      2 | adam  |  
+-----+-----+  
1 row in set (0.00 sec)
```

```
mysql> create database joins;  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> use joins;  
Database changed  
mysql> CREATE TABLE class (  
->     id INT,  
->     name VARCHAR(30)  
-> );  
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> CREATE TABLE class_info (  
->     id INT,  
->     address VARCHAR(30)  
-> );  
Query OK, 0 rows affected (0.05 sec)
```

```
mysql>  
mysql> INSERT INTO class VALUES  
-> (1, 'abhi'),  
-> (2, 'adam'),  
-> (4, 'alex');  
Query OK, 3 rows affected (0.01 sec)  
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql>  
mysql> INSERT INTO class_info VALUES  
-> (1, 'DELHI'),  
-> (2, 'MUMBAI'),  
-> (3, 'CHENNAI');  
Query OK, 3 rows affected (0.01 sec)  
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql>
```

```
MySQL 8.0 Command Line Cli  ×  +  v
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO class_inform VALUES
  -> (1,'DELHI'),
  -> (2,'MUMBAI'),
  -> (3,'CHENNAI');
ERROR 1146 (42S02): Table 'joins.class_inform' doesn't exist
mysql> INSERT INTO class_info VALUES
  -> (1,'DELHI'),
  -> (2,'MUMBAI'),
  -> (3,'CHENNAI');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
+-----+-----+
| id    | name    |
+-----+-----+
| 1     | abhi    |
| 2     | adam    |
| 3     | chester |
+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT name FROM first_table
  -> UNION
  -> SELECT name FROM second_table;
+-----+
| name    |
+-----+
| abhi    |
| adam    |
| chester |
+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM first_table
  -> UNION ALL
  -> SELECT * FROM second_table;
+-----+-----+
| id    | name    |
+-----+-----+
| 1     | abhi    |
| 2     | adam    |
| 2     | adam    |
| 3     | chester |
+-----+-----+
```

```
MySQL 8.0 Command Line Cli  ×  +  ∨  
  
mysql> SELECT c.id, c.name, ci.address  
-> FROM class c  
-> INNER JOIN class_info ci  
-> ON c.id = ci.id  
-> WHERE ci.address IS NOT NULL;
```

id	name	address
1	abhi	DELHI
1	abhi	DELHI
2	adam	MUMBAI
2	adam	MUMBAI
1	abhi	DELHI
1	abhi	DELHI
2	adam	MUMBAI
2	adam	MUMBAI
3	alex	CHENNAI
3	alex	CHENNAI

10 rows in set (0.00 sec)

```
mysql> SELECT c.id, c.name, ci.address  
-> FROM class c  
-> LEFT JOIN class_info ci  
-> ON c.id = ci.id  
-> WHERE ci.address IS NULL;
```

id	name	address
4	alex	NULL
4	anu	NULL
5	ashish	NULL

3 rows in set (0.00 sec)

```
mysql> SELECT COUNT(*)
-> FROM
-> (
-> SELECT * FROM first_table
-> UNION ALL
-> SELECT * FROM second_table
-> ) A;
```

```
+-----+
| COUNT(*) |
+-----+
|         4 |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT c.id,
->         c.name,
->         CASE
->             WHEN ci.address IS NULL THEN 'Address Missing'
->             ELSE 'Address Available'
->         END AS Status
-> FROM class c
-> LEFT JOIN class_info ci
-> ON c.id = ci.id;
```

```
+-----+-----+-----+
| id  | name  | Status          |
+-----+-----+-----+
| 1   | abhi  | Address Available |
| 1   | abhi  | Address Available |
| 2   | adam  | Address Available |
| 2   | adam  | Address Available |
| 4   | alex  | Address Missing   |
| 1   | abhi  | Address Available |
| 1   | abhi  | Address Available |
| 2   | adam  | Address Available |
| 2   | adam  | Address Available |
| 3   | alex  | Address Available |
| 3   | alex  | Address Available |
| 4   | anu   | Address Missing   |
| 5   | ashish | Address Missing   |
+-----+-----+-----+
```

```
13 rows in set (0.00 sec)
```

```
mysql> SELECT c.id, c.name, ci.address
-> FROM class c
-> INNER JOIN class_info ci
-> ON c.id = ci.id
-> WHERE ci.address IS NOT NULL;
```

JOINS_OPERATIONS_DB_11

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```

C Syntax to use here: SELECT name FROM second_table at line 3
mysql> SELECT DISTINCT f.name
      -> FROM first_table f
      -> INNER JOIN second_table s
      -> ON f.name = s.name;

```

```

+-----+
| name |
+-----+
| adam |
+-----+
1 row in set (0.00 sec)

```

```
mysql> SELECT f.*
```

```

mysql> SELECT f.*
      -> FROM first_table f
      -> LEFT JOIN second_table s
      -> ON f.name = s.name
      -> WHERE s.name IS NULL;

```

```

+-----+-----+
| id  | name |
+-----+-----+
|  1  | abhi |
+-----+-----+
1 row in set (0.00 sec)

```

```
mysql>
```

JOINS_OPERATIONS_DB_11

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```
MySQL 8.0 Command Line Cli × + ▾

mysql> SELECT *
-> FROM class
-> RIGHT OUTER JOIN class_info
-> ON class.id=class_info.id;
+-----+-----+-----+-----+
| id | name | id | address |
+-----+-----+-----+-----+
| 1 | abhi | 1 | DELHI |
| 1 | abhi | 1 | DELHI |
| 2 | adam | 2 | MUMBAI |
| 2 | adam | 2 | MUMBAI |
| 3 | alex | 3 | CHENNAI |
| 1 | abhi | 1 | DELHI |
| 1 | abhi | 1 | DELHI |
| 2 | adam | 2 | MUMBAI |
| 2 | adam | 2 | MUMBAI |
| 3 | alex | 3 | CHENNAI |
| NULL | NULL | 7 | NOIDA |
| NULL | NULL | 8 | PANIPAT |
+-----+-----+-----+-----+
12 rows in set (0.00 sec)
```